Lecture 11: Frank–Wolfe and Non-Euclidean Descent

TTIC 31070 / CMSC 35470 / BUSF 36903 / STAT 31015

Convex Optimization

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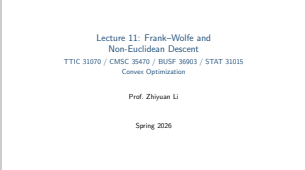

Lectures 7–9 minimized a local quadratic / Bregman model over the feasible set K. That projection / proximal step can be expensive on structured polytopes, spectrahedra, simplices.

Today: Frank–Wolfe replaces the projection oracle with a *linear minimization oracle* (LMO) over K. Cheap when K has structure compatible with linear programming.

From Projection to Linear Minimization

Lectures 7–9 update on K: solve a nonlinear local subproblem over K

$$x_{t+1} \in \operatorname*{argmin}_{x \in K} \left\{ \langle \nabla f(x_t), x - x_t \rangle + \frac{1}{2\eta_t} \|x - x_t\|^2 \right\} \quad \text{or Bregman analog.}$$

For some K (simplex, spectrahedron, ℓ_1 ball, polytopes), even the projection / proximal step is expensive — but *linear minimization* is cheap:

$$s_t \in \operatorname*{argmin}_{s \in K} \langle \nabla f(x_t), s \rangle.$$

Frank–Wolfe (1956). Replace the projection primitive by an LMO; never leave K.

- No norm needed; no symmetric / centered K.
- Compactness $\Rightarrow$ LMO output exists.
- Convex combination keeps iterates in K.

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The shift is conceptual: the old workhorse was projection / Bregman projection. Frank–Wolfe says: only assume linear minimization. That's the cheapest possible nontrivial primitive on K.

For ℓ_1 ball: LMO returns a signed unit basis vector. For spectrahedron $\{X \succeq 0, \operatorname{tr} X = 1\}$: top eigenvector. For simplex: minimum coordinate. Each of these is much cheaper than projection.

Frank–Wolfe Update

Linear minimization oracle (LMO). For $K \subseteq E$ compact convex and $g \in E^*$,

$$\mathrm{LMO}_K(g) := \operatorname*{argmin}_{s \in K} \langle g, s \rangle \quad (\neq \emptyset).$$

Definition 11.1 (Frank–Wolfe update). f differentiable on a neighborhood of K. Given $x_t \in K$, choose $s_t \in \mathrm{LMO}_K(\nabla f(x_t))$ and stepsize $\gamma_t \in [0, 1]$:

$$x_{t+1} := (1 - \gamma_t)\, x_t + \gamma_t\, s_t.$$

Geometry. Linearize f at $x_t \to$ pull toward the LMO atom $s_t \to$ take a convex combination. Iterate stays in K by convexity.

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The picture: at x_t the gradient defines a hyperplane; the LMO picks the most negative-inner-product extreme point of K (a vertex if K is a polytope). The update then takes a small step from x_t toward s_t.

Curvature Constant

Definition 11.2 (FW curvature). $K \subseteq E$ compact convex; f diff'ble. The curvature constant of f over K is

$$C_f := \sup_{\substack{x,s\in K\\ \gamma\in(0,1]}} \frac{2}{\gamma^2}\big(f(x+\gamma(s-x)) - f(x) - \gamma\langle \nabla f(x), s-x\rangle\big).$$

Lemma 11.1 (Quadratic upper model + norm bound). If $C_f < \infty$, then for all $x, s \in K$, $\gamma \in [0,1]$: $f(x+\gamma(s-x)) \le f(x) + \gamma\langle \nabla f(x), s-x\rangle + \frac{\gamma^2}{2} C_f$.
If f is L-smooth w.r.t. $\|\cdot\|$ and $D = \operatorname{diam}_{\|\cdot\|}(K)$: $C_f \le LD^2$ ($\le 4L\rho^2$ if $K = \rho B$).

Why C_f, not LD^2? C_f is intrinsic to (f, K) — norm-free; LD^2 is a lossy norm-dependent upper bound.

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Curvature C_f replaces the role of $L\rho^2$ in unconstrained smooth analysis. It is the smallest constant making the quadratic upper model valid along feasible chords. Choosing a norm gives the upper bound LD^2, but C_f itself doesn't need a norm at all.

Lemma 11.1: Proof

Step 1. Quadratic upper model. For $\gamma \in (0,1]$, the definition of C_f gives

$$\frac{2}{\gamma^2}\big(f(x+\gamma(s-x)) - f(x) - \gamma\langle \nabla f(x), s-x\rangle\big) \le C_f,$$

which rearranges to the displayed inequality. For $\gamma = 0$ both sides equal $f(x)$.

Step 2. Norm-dependent upper bound. If f is L-smooth w.r.t. $\|\cdot\|$, then for $x, s \in K$ and $\gamma \in [0,1]$,

$$f(x+\gamma(s-x)) \le f(x) + \gamma\langle \nabla f(x), s-x\rangle + \tfrac{L}{2}\gamma^2\|s-x\|^2.$$

Since $\|s-x\| \le D$ for $x, s \in K$, taking the sup in the definition gives $C_f \le LD^2$. For $K = \rho B_{\|\cdot\|}$ we have $D = 2\rho$, hence $C_f \le 4L\rho^2$. $\square$

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The first step is purely the definition of C_f. The second step is the only place where smoothness and a norm enter the analysis. After this lemma, the rest of the FW analysis can use either C_f directly (norm-free) or LD^2 (norm-aware).

Frank–Wolfe Gap

Definition 11.3 (FW gap). For $x \in K$,

$$g_{\mathrm{FW}}(x) := \max_{s \in K} \langle \nabla f(x), x - s \rangle = -\min_{s \in K} \langle \nabla f(x), s - x \rangle.$$

Computable from one LMO call. Stationarity certificate: $g_{\mathrm{FW}}(x) = 0 \Leftrightarrow$ first-order optimal.

Theorem 11.2 (Gap upper-bounds suboptimality). If f is convex on K, then for all $x \in K$:

$$f(x) - \min_{y \in K} f(y) \leq g_{\mathrm{FW}}(x).$$

Proof. Convexity at $x^\star \in \operatorname{argmin}_K f$: $f(x^\star) \geq f(x) + \langle \nabla f(x), x^\star - x \rangle$, so $f(x) - f(x^\star) \leq \langle \nabla f(x), x - x^\star \rangle \leq \max_{s \in K} \langle \nabla f(x), x - s \rangle = g_{\mathrm{FW}}(x)$. $\square$

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Two roles for the gap: (1) at the comparator $x^\star \in K$, the gap upper-bounds suboptimality; (2) along the iterate sequence, the gap quantifies the per-step descent because the LMO atom s_t realizes the inner max.

Deterministic Frank–Wolfe Rate

Theorem 11.3 (Frank–Wolfe, 1956; Jaggi, 2013). f convex, $C_f < \infty$, $x_0 \in K$. Run FW with $\gamma_t = \dfrac{2}{t+2}$. Then for all $T \geq 1$,

$$\boxed{f(x_T) - \min_{x \in K} f(x) \leq \frac{2C_f}{T+2}.}$$

Takeaway. $O(1/T)$ rate without projection or norm. With L-smoothness w.r.t. a norm: $\leq 2LD^2/(T+2)$.

Comparison.

- Projected GD (smooth convex): $O(LD^2/T)$ — same rate, but needs projection.
- For sparse / low-rank optimization: each FW iterate is a sparse / rank-1 atom $\Rightarrow$ structured solution.

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The $O(1/T)$ rate is the standard guarantee for general FW with the $\gamma_t = 2/(t+2)$ schedule. Sharper rates exist for special classes (e.g., on polytopes with extra assumptions), but the basic FW theorem stops here.

Thm 11.3: Proof

Step 1. One-step recursion. Let $\delta_t := f(x_t) - f^\star$. Lemma 11.1 + LMO + Thm 11.2:

$$f(x_{t+1}) \le f(x_t) + \gamma_t \langle \nabla f(x_t), s_t - x_t \rangle + \frac{\gamma_t^2}{2} C_f = f(x_t) - \gamma_t g_{\mathrm{FW}}(x_t) + \frac{\gamma_t^2}{2} C_f.$$

Since $g_{\mathrm{FW}}(x_t) \ge \delta_t$, $\delta_{t+1} \le (1-\gamma_t)\delta_t + \frac{\gamma_t^2}{2} C_f$.

Step 2. Induction on $t \ge 1$ that $\delta_t \le 2C_f/(t+2)$. Base $t = 1$ ($\gamma_0 = 1$, $x_1 = s_0$, LMO + Lemma 11.1): $\delta_1 \le \frac{1}{2} C_f \le 2C_f/3$.

Step 3. Inductive step. With $\gamma_t = 2/(t+2)$:

$$\delta_{t+1} \le \left(1 - \frac{2}{t+2}\right) \frac{2C_f}{t+2} + \frac{1}{2}\left(\frac{2}{t+2}\right)^2 C_f = \frac{2C_f(t+1)}{(t+2)^2} \le \frac{2C_f}{t+3},$$

using $(t+1)(t+3) \le (t+2)^2$. $\square$

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The decreasing schedule $\gamma_t = 2/(t+2)$ is critical. Larger steps inflate the curvature term; smaller steps don't kill the suboptimality fast enough. The product γ_t^2 vs $\gamma_t \delta_t$ tradeoff sets the schedule.

Constant-stepsize Frank–Wolfe

Theorem 11.4 (Constant-step FW). Same setup; $\gamma_t \equiv \gamma \in (0,1]$. Then for all $T \geq 0$,

$$\delta_T \leq (1-\gamma)^T \delta_0 + \frac{\gamma C_f}{2}\left(1-(1-\gamma)^T\right) \leq (1-\gamma)^T \delta_0 + \frac{\gamma C_f}{2}.$$

Two regimes.

- *Transient:* $(1-\gamma)^T \delta_0$ — geometric in T, but only if γ large.
- *Error:* $\gamma C_f/2$ — doesn't go to 0 for fixed $\gamma > 0$.

With $\gamma_T = \log T/T$: transient $\leq 1/T$ and error $\leq C_f \log T/(2T)$, so

$$f(x_T) - f^\star \leq \frac{f(x_0)-f^\star}{T} + \frac{C_f \log T}{2T}.$$

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With $\gamma_T = \log T/T$: transient $\leq 1/T$ and error $\leq C_f \log T/(2T)$, so
$f(x_T) - f^\star \leq \frac{f(x_0)-f^\star}{T} + \frac{C_f \log T}{2T}$.

Constant stepsize is the version closer to deep-learning practice. The horizon-dependent $\gamma = \log T/T$ recovers $\tilde{O}(1/T)$ rate. Without horizon knowledge, a decreasing schedule like $2/(t+2)$ is preferred.

Thm 11.4: Proof

Step 1. Same one-step recursion as Thm 11.3. With $\gamma_t \equiv \gamma$,

$$\delta_{t+1} \le (1-\gamma)\delta_t + \frac{\gamma^2}{2} C_f.$$

Step 2. Unroll the affine recursion. Iterating,

$$\delta_T \le (1-\gamma)^T \delta_0 + \frac{\gamma^2}{2} C_f \sum_{k=0}^{T-1} (1-\gamma)^k = (1-\gamma)^T \delta_0 + \frac{\gamma C_f}{2}\big(1-(1-\gamma)^T\big).$$

The looser form $\le (1-\gamma)^T \delta_0 + \gamma C_f/2$ drops the $(1-(1-\gamma)^T)$ factor.

Step 3. Horizon-only choice. With $\gamma_T = \log T / T$ $(T \ge 2)$,

$$(1-\gamma_T)^T \le \exp(-\gamma_T T) = \tfrac{1}{T}.$$

Substituting into the transient-plus-error bound proves $f(x_T) - f^\star \le \frac{\delta_0}{T} + \frac{C_f \log T}{2T}$. $\square$

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Step 2 is just a geometric series. Step 3 picks γ in terms of horizon T to balance the two regimes.

Normalized Steepest Descent with Weight Decay

Recall. Steepest descent w.r.t. norm $\|\cdot\|$ uses $v_t \in \mathrm{LMO}_B(\nabla f(x_t))$ where $B = \{v : \|v\| \le 1\}$.

Definition 11.4 (NSD with decoupled weight decay). Norm $\|\cdot\|$, weight decay $\lambda > 0$, learning rate $\eta_t \in [0, 1/\lambda]$. Choose $v_t \in \mathrm{LMO}_B(\nabla f(x_t))$ and update

$$\boxed{x_{t+1} = (1 - \lambda\eta_t)\, x_t + \eta_t\, v_t.}$$

Takeaway.

- $(1 - \lambda\eta_t)x_t$: shrink the iterate toward 0 (*weight decay*).
- $\eta_t v_t$: take a normalized steepest-descent step.
- Decoupled: λ shrinks x_t, not $\nabla f(x_t)$ (different from ℓ_2 regularization!).

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Decoupled weight decay shrinks the parameter directly each step; ℓ_2 regularization adds λx_t to the gradient. They are NOT equivalent in general (e.g., once any non-trivial preconditioning is involved).

NSD with Weight Decay = Frank–Wolfe on a Norm Ball

Proposition 11.6. Let $B = \{\|v\| \le 1\}$, $K = \frac{1}{\lambda}B$. Then NSD with decoupled weight decay is exactly Frank–Wolfe on K, with

$$\gamma_t = \lambda\eta_t, \qquad s_t = \tfrac{1}{\lambda}v_t \in K.$$

Conversely, FW on $K = \rho B$ with stepsize γ_t is NSD-WD with $\lambda = 1/\rho$, $\eta_t = \rho\gamma_t$.

Proof. $\mathrm{LMO}_K = \frac{1}{\lambda}\mathrm{LMO}_B$, so $s_t = v_t/\lambda$. The FW update with $\gamma_t = \lambda\eta_t$ is $(1-\gamma_t)x_t + \gamma_t s_t = (1-\lambda\eta_t)x_t + \eta_t v_t$. $\square$

Consequence (Thm 11.4 with $C_f \le 4L/\lambda^2$):

$$f(x_T) - \min_{\|x\|\le 1/\lambda} f(x) \le (1-\lambda\eta)^T\delta_0 + \frac{2L\eta}{\lambda}.$$

With $\eta_T = \log T/(\lambda T)$: $\delta_0/T + 2L\log T/(\lambda^2 T)$.

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With $\eta_T = \log T/(\lambda T)$: $\delta_0/T + 2L\log T/(\lambda^2 T)$.

This is the bridge from FW theory to modern optimizer practice. Decoupled weight decay is not a heuristic; it's a mathematical equivalent of running FW on a norm ball. The "norm ball constraint" interpretation explains why weight decay regularizes the solution norm.

Why Weight Decay Matters: Radius Comparison

NSD with weight decay (FW on $K = B/\lambda$):

$$f(x_T) - \min_{\|x\| \le 1/\lambda} f(x) \le \frac{2L/\lambda^2}{T+2}.$$

Comparator radius $= 1/\lambda$ (chosen by us).

NSD without weight decay (assume monotone iterates stay in initial sublevel set):

$$f(x_T) - f(x^\star) \le \frac{2LR_{\mathrm{sub}}^2}{T+2}, \qquad R_{\mathrm{sub}} := \sup\{\|x - x^\star\| : f(x) \le f(x_0)\}.$$

Both look like $O(LR^2/T)$. But the radius means different things:

- With WD: $R = 1/\lambda$ is a *chosen* feasible radius; comparator is the constrained min.
- Without WD: $R = R_{\mathrm{sub}}$ is the radius of the *initial sublevel set* around $x^\star$.

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- With WD: $R = 1/\lambda$ is a *chosen* feasible radius; comparator is the constrained min.
- Without WD: $R = R_{\mathrm{sub}}$ is the radius of the *initial sublevel set* around $x^\star$.
R_{sub} can be arbitrarily large, even when a useful centered ball exists.

The statements look superficially similar but they are different theorems. The without-WD bound has nothing to choose: the sublevel-set radius is what it is. The with-WD bound lets us tune the constraint radius $1/\lambda$ to balance optimization vs. regularization.

Example: Sublevel Radius Can Be Huge

Example 11.1. Asymmetric quadratic on $\mathbb{R}$: for $0 < \varepsilon \le 1$,

$$f_\varepsilon(x) = \begin{cases} \frac{1}{2}(x-1)^2, & x \le 1, \\ \frac{\varepsilon}{2}(x-1)^2, & x \ge 1. \end{cases}$$

1-smooth; $x^\star = 1$.

Starting at $x_0 = 0$**:** $f_\varepsilon(0) = 1/2$. The sublevel set is

$$\{x : f_\varepsilon(x) \le 1/2\} = [0,\, 1 + 1/\sqrt{\varepsilon}],$$

so $R_{\mathrm{sub}} = 1/\sqrt{\varepsilon}$ — arbitrarily large as $\varepsilon \to 0$.

But the centered domain $K = [-1, 1]$ already contains $x^\star = 1$ and has radius 1.

Takeaway. The two statements measure different geometric quantities; you cannot just compare the constants.

Example: Sublevel Radius Can Be Huge
Example 11.1. Asymmetric quadratic on $\mathbb{R}$: for $0 < \varepsilon \le 1$,
$$f_\varepsilon(x) = \begin{cases} \frac{1}{2}(x-1)^2, & x \le 1, \\ \frac{\varepsilon}{2}(x-1)^2, & x \ge 1. \end{cases}$$
1-smooth; $x^\star = 1$.
Starting at $x_0 = 0$: $f_\varepsilon(0) = 1/2$. The sublevel set is
$$\{x : f_\varepsilon(x) \le 1/2\} = [0,\, 1 + 1/\sqrt{\varepsilon}],$$
so $R_{\mathrm{sub}} = 1/\sqrt{\varepsilon}$ — arbitrarily large as $\varepsilon \to 0$.
But the centered domain $K = [-1, 1]$ already contains $x^\star = 1$ and has radius 1.
Takeaway. The two statements measure different geometric quantities; you cannot just compare the constants.

The example is one-dimensional but the issue is real in high dimensions: shallow / asymmetric loss surfaces yield enormous sublevel sets even when the relevant scale of the solution is moderate. Adding weight decay imposes a controlled radius; it's not a free regularizer, it's a geometric reset.

LMO Atoms on Common Norm Balls

Norm ball B	**LMO atom** $\mathrm{LMO}_B(g)$
ℓ_2	$-g/\|g\|_2$
ℓ_∞	$-\operatorname{sign}(g)$ entrywise
ℓ_1	$-\operatorname{sign}(g_{i^\star})e_{i^\star}$, where $i^\star \in \operatorname{argmax}_i \lvert g_i \rvert$
spectral norm ball	$-UV^\top$ from $G = U\Sigma V^\top$
nuclear norm ball	top rank-one atom $-uv^\top$
spectrahedron	$-vv^\top$, where v is a top eigenvector of G

Optimizer connection. Each row gives the LMO direction. Frank–Wolfe on that norm ball is equivalently a normalized update with decoupled weight decay.

ℓ_∞ ball + FW/LMO = sign update + WD (Lion-style). Spectral ball + FW/LMO = orthogonalized momentum update + WD (Muon-style).

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ℓ_∞ ball + FW/LMO = sign update + WD (Lion-style). Spectral ball + FW/LMO = orthogonalized momentum update + WD (Muon-style).

This table should be an atom catalog, not a crowded optimizer taxonomy. The unifying perspective: pick a norm, compute the polar atom through the LMO, and then read FW on the corresponding ball as a weight-decayed normalized update.

Naive Stochastic Frank–Wolfe

Definition 11.5 (Naive single-sample stochastic FW). Filtration $\mathcal{F}_t$; x_t is $\mathcal{F}_t$-measurable; unbiased estimator $\widehat{g}_t$:

$$\mathbb{E}[\widehat{g}_t \mid \mathcal{F}_t] = \nabla f(x_t), \qquad s_t \in \mathrm{LMO}_K(\widehat{g}_t), \qquad x_{t+1} = (1-\gamma_t)x_t + \gamma_t s_t.$$

Question. Does the deterministic $O(C_f/T)$ rate carry over with σ-bounded noise?

Answer. No. Naive stochastic FW has a *non-vanishing error*, regardless of how small the learning rate is and how large the number of steps T is.

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Question. Does the deterministic $O(C_f/T)$ rate carry over with σ-bounded noise?
Answer. No. Naive stochastic FW has a non-vanishing error, regardless of how small the learning rate is and how large the number of steps T is.

The question is natural because mirror descent and SGD both tolerate unbiased noise. Frank–Wolfe is different because the LMO is a nonlinear map applied to the noisy gradient. We pair the noise with the atom that the noise itself selected — and that pairing is biased.

Stochastic FW: Non-vanishing Error

Theorem 11.7 (Non-vanishing error). f convex, $C_f < \infty$, $D = \operatorname{diam}_{\|\cdot\|}(K)$, $\mathbb{E}[\|\widehat{g}_t - \nabla f(x_t)\|_* \mid \mathcal{F}_t] \le \sigma$. Run naive stochastic FW with $\gamma_t = 2/(t+2)$. Then

$$\boxed{\mathbb{E}[f(x_T) - \min_K f] \le \frac{2C_f}{T+2} + D\sigma.}$$

The $D\sigma$ error does not vanish as $T \to \infty$.

Takeaway. The $2C_f/(T+2)$ piece is the deterministic rate; the additive $D\sigma$ piece is a non-vanishing error that grows linearly in σ and in the diameter D, and is independent of T.

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Takeaway. The $2C_f/(T+2)$ piece is the deterministic rate; the additive $D\sigma$ piece is a non-vanishing error that grows linearly in σ and in the diameter D, and is independent of T.

Compare Thm 11.3: deterministic FW gets $2C_f/(T+2)$ with no such error. Naive stochastic FW with the same stepsize gets the same deterministic piece plus an error we cannot drive down.

Thm 11.7: Proof (1/2)

Let $x^\star \in \operatorname{argmin}_K f$, $\delta_t := f(x_t) - f(x^\star)$.

Step 1. Curvature inequality (Lem 11.1).

$$\delta_{t+1} \le \delta_t + \gamma_t \langle \nabla f(x_t), s_t - x_t \rangle + \frac{\gamma_t^2}{2} C_f.$$

Step 2. Insert/subtract $\widehat{g}_t$, use LMO on $\widehat{g}_t$. $\langle \widehat{g}_t, s_t - x_t \rangle \le \langle \widehat{g}_t, x^\star - x_t \rangle$ since $x^\star \in K$. Thus

$$\langle \nabla f(x_t), s_t - x_t \rangle = \langle \widehat{g}_t, s_t - x_t \rangle + \langle \nabla f(x_t) - \widehat{g}_t, s_t - x_t \rangle.$$

Add and subtract $\langle \widehat{g}_t, x^\star - x_t \rangle$:

$$\langle \nabla f(x_t), s_t - x_t \rangle \le \langle \nabla f(x_t), x^\star - x_t \rangle + \langle \widehat{g}_t - \nabla f(x_t), x^\star - s_t \rangle.$$

Step 3. Convexity. $\langle \nabla f(x_t), x^\star - x_t \rangle \le -\delta_t$. Combine with Step 1:

$$\delta_{t+1} \le (1 - \gamma_t)\delta_t + \frac{\gamma_t^2}{2} C_f + \gamma_t \langle \widehat{g}_t - \nabla f(x_t), x^\star - s_t \rangle.$$

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The crucial line is the third one: by adding and subtracting, we move the noise into a pairing $\langle \xi_t, x^\star - s_t \rangle$ where s_t depends on ξ_t. This is what blocks vanishing of the noise term in expectation.

Thm 11.7: Proof (2/2)

Step 4. Hölder + bounded diameter. $\|x^\star - s_t\| \le D$, so

$$|\langle \widehat{g}_t - \nabla f(x_t), x^\star - s_t\rangle| \le \|\widehat{g}_t - \nabla f(x_t)\|_* \cdot D.$$

Conditional expectation gives $\mathbb{E}[|\cdots| \mid \mathcal{F}_t] \le D\sigma$. (Note: *not* 0, because s_t depends on $\widehat{g}_t$.)

Step 5. Take expectation, recurse. $\mathbb{E}\delta_{t+1} \le (1-\gamma_t)\mathbb{E}\delta_t + \frac{\gamma_t^2}{2}C_f + \gamma_t D\sigma$.

Step 6. Induction with $\gamma_t = 2/(t+2)$. Same argument as Thm 11.3 with the additional constant $D\sigma$:

$$\mathbb{E}\delta_T \le \frac{2C_f}{T+2} + D\sigma.$$

Base case $\mathbb{E}\delta_1 \le \frac{1}{2}C_f + D\sigma \le 2C_f/3 + D\sigma$. $\square$

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The induction in Step 6 is exactly the same as for the deterministic theorem, except the recurrence has an extra additive $\gamma_t D\sigma$ term. Because $\sum_t \gamma_t = \infty$, the noise term doesn't telescope — it persists as a constant error.

Constant-step Stochastic FW Error

Corollary 11.8 (Constant-step stochastic error). Same setup as Thm 11.7; $\gamma_t \equiv \gamma \in (0, 1]$. Then

$$\mathbb{E}\delta_T \le (1-\gamma)^T \delta_0 + \left(D\sigma + \frac{\gamma C_f}{2}\right)\left(1 - (1-\gamma)^T\right) \le (1-\gamma)^T \delta_0 + D\sigma + \frac{\gamma C_f}{2}.$$

Proof. From the Step-5 recursion of Thm 11.7 with $\gamma_t \equiv \gamma$:

$$\mathbb{E}\delta_{t+1} \le (1-\gamma)\mathbb{E}\delta_t + \gamma D\sigma + \frac{\gamma^2}{2} C_f.$$

Unrolling this affine recursion gives the displayed bound. □

Takeaway. Compared with deterministic Thm 11.4: the stochastic error adds $D\sigma$ on top of $\gamma C_f/2$. Even at $\gamma \to 0$, the $D\sigma$ persists.

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Takeaway. Compared with deterministic Thm 11.4: the stochastic error adds $D\sigma$ on top of $\gamma C_f/2$. Even at $\gamma \to 0$, the $D\sigma$ persists.

The stochastic error is doubly bad: even after running long enough that the transient term $(1-\gamma)^T \delta_0$ vanishes, you still pay $D\sigma + \gamma C_f/2$. Shrinking γ helps the curvature piece but the noise piece is independent of γ.

A Non-vanishing Stochastic Error

The proof of Thm 11.7 gives the one-step recursion

$$\mathbb{E}[\delta_{t+1} \mid \mathcal{F}_t] \le (1-\gamma_t)\delta_t + \frac{\gamma_t^2}{2} C_f + \gamma_t D\sigma, \qquad \delta_t = f(x_t) - f(x^\star). \tag{FW}$$

The key scaling. In one step,

$$\underbrace{-\gamma_t \delta_t}_{\text{useful progress}} \quad + \quad \underbrace{\gamma_t D\sigma}_{\text{biased LMO noise}} \quad + \quad \underbrace{\frac{\gamma_t^2}{2} C_f}_{\text{curvature}} .$$

The curvature term is smaller order when $\gamma_t \to 0$; the biased noise term is not.

Let $S_T = \sum_{t<T} \gamma_t$. Normalizing the accumulated noise by the total movement gives

$$\frac{1}{S_T} \sum_{t<T} \gamma_t D\sigma = D\sigma.$$

So even if $\gamma_t \to 0$ and $S_T \to \infty$, the error created by feeding a single noisy gradient (covector) into the LMO does *not* average away.

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This is the conceptual point students need to see: the problem is not that the stepsize is too large. The useful movement and the stochastic LMO bias both carry the same coefficient γ_t. Taking γ_t to zero removes the curvature term, but it does not remove the $D\sigma$ term after normalization by the total amount of movement.

MD: Constant LR Averages Gradients

Let h be the mirror map and set the dual coordinate $z_t = \nabla h(x_t)$. With a constant learning rate η, MD with $\widehat{g}_t = g_t + \xi_t$ is

$$z_{t+1} = z_t - \eta \widehat{g}_t, \qquad x_{t+1} = \nabla h^*(z_{t+1}).$$

Unroll and divide by the total movement ηT:

$$\frac{z_1 - z_{T+1}}{\eta T} = \frac{1}{T} \sum_{t<T} g_t + \frac{1}{T} \sum_{t<T} \xi_t.$$

Thus many small MD steps behave like averaging noisy gradients (covectors) before the mirror map sends the result back to primal space.

FW does the opposite. It sends the noisy gradient (covector) through a nonlinear atom selector before averaging:

$$s_t \in \mathrm{LMO}_K(g_t + \xi_t), \qquad \mathbb{E}[\mathrm{LMO}_K(g + \xi)] \neq \mathrm{LMO}_K(g) \text{ in general.}$$

Small steps average selected atoms, not the gradients (covectors) themselves.

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Small steps average selected atoms, not the gradients (covectors) themselves.

This is the missing mechanism. With constant learning rate, the dual coordinate displacement is proportional to the average noisy gradient (covector). Frank–Wolfe instead applies a nonlinear, often discontinuous, atom-selection map to the noisy gradient (covector) at each step.

Constant-LR MD: The Bound

Why does the averaging picture enter the proof? Since x_t and the comparator u are chosen before seeing the current noise,

$$\mathbb{E}[\langle \xi_t, x_t - u\rangle \mid \mathcal{F}_t] = 0.$$

The noise appears only through a predictable linear pairing.

With constant η and $\bar{x}_T = T^{-1}\sum_{t<T} x_t$, a typical MD bound is

$$\mathbb{E}[f(\bar{x}_T) - f(u)] \le \frac{D_h(u, x_1)}{\eta T} + \frac{\eta}{2\alpha}(G^2 + \sigma^2), \qquad \mathbb{E}\|\widehat{g}_t\|_*^2 \le G^2 + \sigma^2.$$

Small LR + many steps. For a fixed horizon, choose one constant $\eta \asymp T^{-1/2}$. Then both terms are $O(T^{-1/2})$. Naive stochastic FW has the extra $D\sigma$ term instead: the noise has already selected the atom before any averaging occurs.

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Small LR + many steps. For a fixed horizon, choose one constant $\eta \asymp T^{-1/2}$. Then both terms are $O(T^{-1/2})$. Naive stochastic FW has the extra $D\sigma$ term instead: the noise has already selected the atom before any averaging occurs.

Do not say mirror descent has no noise problem. It pays the usual second-moment term. The simplified constant-learning-rate message is: choose one small eta for the horizon; the initial term is $1/(\eta T)$ and the stochastic price is $O(\eta)$. In naive stochastic FW, the current noise has already selected the atom, so the bias cannot be removed by taking many small steps.

Large-Batch FW: Denoise Before the LMO

Principle. FW cannot rely on many small steps to average gradients (covectors). Average the stochastic gradients *before* the LMO:

$$\bar{g}_t = \frac{1}{B}\sum_{b=1}^{B} g(x_t;\xi_{t,b}), \qquad s_t \in \mathrm{LMO}_K(\bar{g}_t), \qquad x_{t+1} = (1-\gamma_t)x_t + \gamma_t s_t.$$

If

$$\mathbb{E}[\|\bar{g}_t - \nabla f(x_t)\|_* \mid \mathcal{F}_t] \le \sigma_B,$$

then the proof of Thm 11.7 applies with σ replaced by σ_B. The non-vanishing residual term becomes $D\sigma_B$.

Caveat. In Euclidean norm, independent averaging often gives $\sigma_B \asymp B^{-1/2}$. For a general norm, this rate is an additional concentration assumption, not automatic.

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This slide should now follow the notes: first state the algorithm, then explain how it enters the theorem. Avoid a generic table of noise scales because the large-batch statement is theorem-backed only after an effective-noise assumption, and the scaling is norm-dependent.

Momentum FW: Cheap Denoising

Algorithmic template. Keep a gradient (covector) memory and query the LMO using that memory, with $m_{-1} = 0$:

$$m_t = \beta_t m_{t-1} + (1 - \beta_t)\widehat{g}_t, \qquad s_t \in \mathrm{LMO}_K(m_t),$$

$$x_{t+1} = (1 - \gamma_t)x_t + \gamma_t s_t.$$

What changes? Momentum can reduce variance cheaply, but it is biased: past gradients were evaluated at past points. Smoothness and slow movement are needed to control this bias.

Known result. In the Hilbert/Euclidean stochastic FW setting, the survey of Braun et al. summarizes an $O(T^{-1/3})$ momentum SFW rate under smoothness, compact diameter, and bounded second-moment noise.

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This is now algorithm-first, matching the notes. The key distinction from large batch: momentum is a practical cheap denoising template. The known theorem is Hilbert/Euclidean and uses gradient-tracking estimates, so do not present it as a general norm-space theorem here.

Two-Parameter Momentum Optimizers

Practical sign / orthogonalized optimizers often use a Lion-style two-beta momentum template: one gradient (covector) chooses the current LMO direction, while another persistent memory is carried forward.

$$c_t = \beta_1 m_{t-1} + (1-\beta_1)\widehat{g}_t, \qquad m_t = \beta_2 m_{t-1} + (1-\beta_2)\widehat{g}_t,$$

$$v_t \in \mathrm{LMO}_B(c_t), \qquad x_{t+1} = (1-\lambda_t)x_t + \eta_t v_t.$$

- **Lion**: B is the ℓ_∞ ball, so $v_t = -\operatorname{sign}(c_t)$.
- **Muon**: B is the spectral norm ball, so $v_t = -UV^\top$ from $C_t = U\Sigma V^\top$.

This is the optimizer template. The lecture theorem explains why the gradient (covector) must be denoised before the LMO; it does not fully analyze this two-parameter momentum heuristic.

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Separate the notes-backed single-filter momentum FW template from the practical two-parameter optimizer template. This avoids pretending the theorem analyzes Lion or Muon.

Summary

Frank–Wolfe replaces projection by linear minimization.

- Norm-free, no symmetry, no centered ball.
- Curvature constant C_f is the intrinsic quantity; LD^2 is a norm-dependent upper bound.
- Decreasing-stepsize $\gamma_t = 2/(t+2)$ gives $O(C_f/T)$.

Norm-ball specialization.

- FW on norm ball $\Leftrightarrow$ NSD with decoupled weight decay.
- Atom catalog: $\ell_\infty \to$sign, spectral$\to$polar, $\ell_1 \to$basis, ...
- Modern optimizers (Lion, Muon) are momentum-FW on a chosen norm ball.

Stochastic warning. MD averages noisy gradients (covectors) in dual coordinates. Naive stochastic FW first selects an atom from the noisy gradient (covector), creating a non-vanishing $D\sigma$ error. Remedy: denoise before the LMO (large batch or momentum).

Next. Lower bounds for first-order methods.

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Atom catalog: $\ell_\infty \to$sign, spectral$\to$polar, $\ell_1 \to$basis, ...
Modern optimizers (Lion, Muon) are momentum-FW on a chosen norm ball.
Stochastic warning. MD averages noisy gradients (covectors) in dual coordinates. Naive stochastic FW first selects an atom from the noisy gradient (covector), creating a non-vanishing $D\sigma$ error. Remedy: denoise before the LMO (large batch or momentum).
Next. Lower bounds for first-order methods.

Lecture 11 is the bridge between classical convex-optimization theory and the modern optimizer landscape. The thread: norm $\to$ atom $\to$ FW + weight decay $\to$ optimizer.

Bibliographic Notes

- **Frank & Wolfe (Naval Res. Logistics Quart. 1956).** Quadratic programming.
- **Demyanov & Rubinov (1967).** Conditional-gradient method, smooth functional.
- **Jaggi (ICML 2013).** Modern projection-free / dual-gap viewpoint, $O(1/T)$ rate.
- **Hazan & Luo (ICML 2016); Mokhtari, Hassani & Karbasi (JMLR 2020).** Variance-reduced / momentum stochastic FW.
- **Bernstein & Newhouse (2024).** Norm-ball / polar duality view of modern optimizers.
- **Lion (Chen et al. 2023); Muon (Jordan et al. 2024; Liu et al. 2025).** Sign / spectral norm-ball directions in deep learning.
- **Type-2 / 2-smooth norms.** Ledoux–Talagrand (1991); Pinelis (1994); Juditsky–Nemirovski (2008).

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- **Frank & Wolfe (Naval Res. Logistics Quart. 1956).** Quadratic programming.
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Frank–Wolfe was originally formulated for quadratic programming. The modern projection-free framing and curvature-constant analysis are due to Jaggi (2013). The bridge to modern deep-learning optimizers (Lion, Muon) is recent (Bernstein–Newhouse 2024 and follow-ups).